

**Assessing the Effects of 2-in-1 Organic Fertilizer and Insecticide on Tomato
(*Solanum lycopersicum*) in Hagonoy, Bulacan**

SAN PEDRO NATIONAL HIGH NATIONAL SCHOOL
SENIOR HIGH SCHOOL DEPARTMENT

**Assessing the Effects of 2-in-1 Organic Fertilizer and Insecticide on Tomato
(*Solanum lycopersicum*) in Hagonoy, Bulacan**

Allanic, DJ Joseph DR.

Medina, Joseph Ivan DL.

Panganiban, Ronald Clement R.

Pineda, Paolo Gabriel N.

Sotto, Roselle Anne J.

San Pedro National High School

Senior High School Department

Science, Technology, Engineering, and Mathematics

2023-2024



Assessing the Effects of 2-in-1 Organic Fertilizer and Insecticide on Tomato (*Solanum lycopersicum*) in Hagonoy, Bulacan

SAN PEDRO NATIONAL HIGH NATIONAL SCHOOL
SENIOR HIGH SCHOOL DEPARTMENT

Abstract

The study conducted in Hagonoy, Bulacan addresses numerous challenges encountered in tomato cultivation, aiming to assist local farmers in enhancing their practices. By developing 3 organic solutions with various formulations, the researchers aim to tackle issues including nutrient deficiencies and pest infestations typically faced inside the region. These solutions were applied to tomato plants

and compared with untreated plants and those treated with traditional chemical products. Over a three-week length, data on plant growth, pest presence, and leaf coloration had been collected and analyzed

the usage of statistical techniques which includes analysis of Variance (ANOVA). The results demonstrated significant improvements in plant growth and effective pest control with the organic solutions, particularly highlighting the efficacy of a 2-in-1 organic fertilizer and insecticide. This innovative approach gives a promising avenue towards sustainable and efficient farming practices in tomato cultivation, addressing the agricultural challenges in Hagonoy, Bulacan.

Tomatoes are a staple crop in Hagonoy, Bulacan. However, it faces many challenges including nutrient deficiencies, soil-related issues, and the need for pest control. The study by Gao, Chen, Gao, Wang, and Qin (2014) titled "Multifunctional Insecticide Fertilizer for Tomatoes and Preparation Method of Multifunctional Insecticide Fertilizer" presents a unique approach to address these challenges, developing a multifunctional insecticide fertilizer.

Gao et al. (2014) designed a multifunctional insecticide fertilizer to enhance nutrient content, soil quality, and effective pest control simultaneously. This research presents potential solutions

to the agricultural problems in Hagonoy, Bulacan and the surrounding areas. By synthesizing a multifunctional fertilizer, it offers insights that can be implemented to enhance tomato cultivation.



Assessing the Effects of 2-in-1 Organic Fertilizer and Insecticide on Tomato (*Solanum lycopersicum*) in Hagonoy, Bulacan

SAN PEDRO NATIONAL HIGH NATIONAL SCHOOL SENIOR HIGH SCHOOL DEPARTMENT

Since Hagonoy, Bulacan faces unique environmental conditions, it may require adjustments to the methods of Gao et al. (2014). Thus, an empirical assessment of the effects of organic fertilizer and insecticide in this region is essential to determine the practicality and adaptability of these methods. This study seeks to provide insights into the potential effects of adopting a multifunctional approach to tomato cultivation in Hagonoy, Bulacan through this research.

The researchers identified pest infestation as a potential reason for the limited cultivation of tomato plants in their area. They recognized that integrating an insecticide into fertilizer could be a solution to ease planting difficulties and protect plants from withering and pest damage.

Pest infestations are a common challenge, prompting the development of a 2-in-1 fertilizer and insecticide solution. However, chemical alternatives pose logistical challenges, high expenses, and environmental concerns for both farmers and consumers. In Hagonoy, tomato cultivation lags behind fishing, possibly due to insufficient pest management options. Embracing organic fertilizer not only simplifies planting but also encourages tomato farming as a viable alternative income stream in Hagonoy, diversifying the local economy beyond fishing.

This proposed research project is based on the need to develop practices for tomato cultivation in Hagonoy, Bulacan. Drawing on the findings of Gao et al. (2014), this research aims to assess the multifunctional insecticide fertilizers in the local context.

Tomato cultivation is a key agricultural practice in Hagonoy, Bulacan, contributing to the region's agricultural economy and local cuisine, affecting the livelihoods of numerous farmers. According to the Local Agricultural Office data (2023), tomato cultivation accounts for approximately 40% of the total agricultural output and estimated 2,000 hectares of land for tomato cultivation Hagonoy, Bulacan, emphasizing its economic importance and impact on local farming practices.



Assessing the Effects of 2-in-1 Organic Fertilizer and Insecticide on Tomato (*Solanum lycopersicum*) in Hagonoy, Bulacan

SAN PEDRO NATIONAL HIGH SCHOOL SENIOR HIGH SCHOOL DEPARTMENT

However, the region faces challenges in tomato cultivation. This includes climatic fluctuations, soil degradation, and pest infestations which hinders successful cultivation, production volume and also compromises the quality and economic yield of the crops. Moreover, the choice of fertilization methods, particularly between organic and inorganic solutions, is crucial to farmers in choosing the approach to enhance tomato yield while ensuring the production of healthy and high-quality crops. Furthermore, the research problem in Hagonoy, Bulacan, revolves around the challenges encountered in the cultivation of tomatoes. This study focuses on the impact of organic fertilizers and insecticides on improving tomato quality, production, and growth.

An assemblage of pivotal studies contributes significantly to elucidating the variables and terms pertinent to the impact of organic fertilizers and insecticides on tomato cultivation.

Firstly, the literature broadly defines organic fertilizers as naturally occurring substances derived from living organisms, such as animal manure, compost, or plant-based materials. These fertilizers are known to enhance soil fertility and plant growth while maintaining ecological balance (Conde & De Asis, 2021; Gorme et al., 2017). Secondly, insecticides are chemical formulations designed to combat insects that pose a threat to crops. Their usage varies widely in agriculture, aiming to control pest populations to prevent damage and losses in crop yield (Gao et al., 2014). Lastly is tomato Yield and Quality, the parameters of tomato yield and quality encompass various aspects, including fruit quantity, size, taste, nutritional content, and overall plant health (Ashraful et al., 2017; Kalbani et al., 2016; Elias et al., 2016). These factors are

crucial indicators of successful tomato cultivation practices. Also, the studies conducted by Kalbani et al. (2016) and Elias et al. (2016) emphasize the effectiveness of specific organic

fertilizers, such as chicken manure, cow manure, and compost, on tomato plant growth. These studies highlight the varied impact of these fertilizers on plant height, fresh and dry weight, and overall fruit quality. The tomato yield and quality metrics is the parameters encompassing fruit



Assessing the Effects of 2-in-1 Organic Fertilizer and Insecticide on Tomato (*Solanum lycopersicum*) in Hagonoy, Bulacan

SAN PEDRO NATIONAL HIGH NATIONAL SCHOOL SENIOR HIGH SCHOOL DEPARTMENT

yield, size, nutritional content, taste, and overall health of the tomato plants.

Ultimately, understanding these terms and variables derived from diverse literature sources lays the groundwork for a comprehensive assessment of the effects of organic fertilizer and insecticide on tomato cultivation in Hagonoy, Bulacan.

The cultivation of tomatoes in Hagonoy, Bulacan, constitutes a vital segment of the local agricultural landscape, yet it grapples with an array of daunting challenges that demand immediate attention and resolution. The Local Agricultural Office's data of Hagonoy (2023) underscores the pivotal role of tomato cultivation, contributing approximately 40% to the total agricultural output in the region. This substantial economic contribution emphasizes its significance as a primary revenue generator and a cornerstone of the local farming economy. Spanning an estimated 2,000 hectares of agricultural land, the vast expanse dedicated to tomato cultivation further highlights its scale and indispensable role within the region's agricultural framework.

The choice of fertilization methods, particularly the deliberation between organic and inorganic solutions, emerges as a critical factor in addressing these challenges. Optimal decisions regarding the selection and application of fertilizers hold immense significance in mitigating adversities and ensuring robust tomato growth.

This research aims to evaluate the effects of organic fertilizer and insecticide use on tomato farming in Hagonoy, Bulacan. Additionally, it aims to support the city's economic growth by creating organic fertilizers that will help improve crop cultivation and provide an alternative source of income. This initiative is a response to a lack of food and rising prices in the city, caused by economic factors. The Local Agricultural Office of Hagonoy emphasizes the agricultural potential of the city and the importance of this research. Hagonoy has 5,000 hectares of farmland that could be utilized for increased productivity. Market analysis shows a 30% increase in food and vegetable prices due to factors like reliance on exports and temporary food



Assessing the Effects of 2-in-1 Organic Fertilizer and Insecticide on Tomato (*Solanum lycopersicum*) in Hagonoy, Bulacan

**SAN PEDRO NATIONAL HIGH SCHOOL
SENIOR HIGH SCHOOL DEPARTMENT**

shortages.

This research focuses on creating 2-in-1 organic fertilizers and pesticides to address global food issues and reduce dependence on expensive imports. This innovation aims to support affordable and sustainable farming practices for local farmers in Hagonoy, helping them generate additional income by growing and selling vegetables. The fundamental of this research is to thoroughly assess the effects of the use of organic fertilizers and insecticides on tomato cultivation in Hagonoy, Bulacan. By carefully analyzing these agricultural interventions, the study aims to understand how they affect the growth, yield, and quality of tomatoes. The findings will provide important insights for sustainable farming practices.

The objectives of this research are to provide practical recommendations for sustainable and productive tomato farming in Hagonoy, Bulacan by studying the effects of organic fertilizer and insecticide use. The aim is to offer empirical insights and evidence-based conclusions to improve tomato growing practices in the region. This research aims to study the impact of using organic fertilizer and insecticide on tomato farming and also suggests a new approach to support local agriculture. By creating affordable organic fertilizers and pesticides made locally, the study aims to reduce the burden of increasing food costs, promote economic stability, and offer extra income opportunities for the people of Hagonoy.

